Computing Methods for Experimental Physics and Data Analysis

Data Analysis in Medical Physics

Lecture 2: intro to MATLAB, read and display medical images

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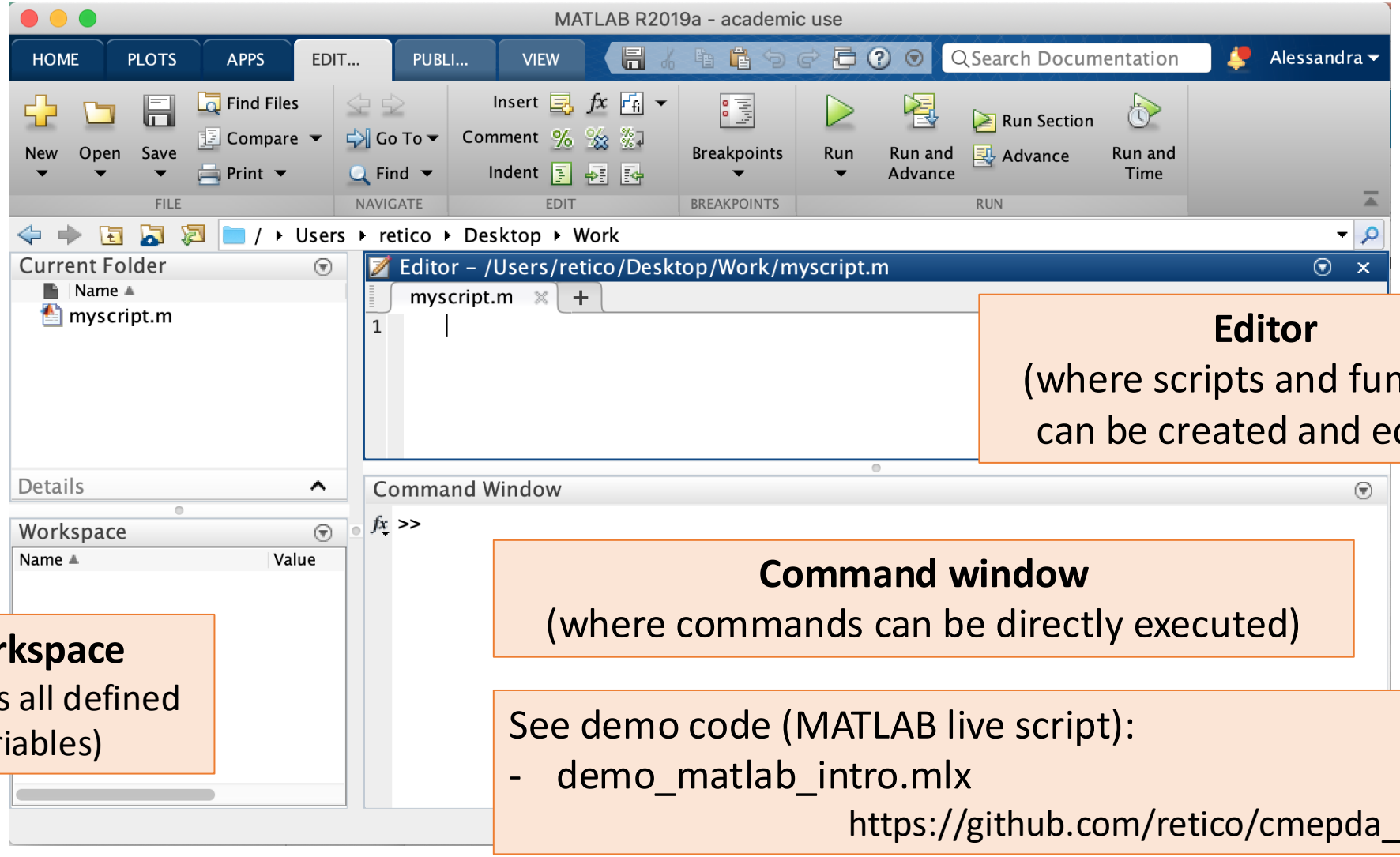
Brief introduction to MATLAB

- MATLAB (MATrix LABoratory) integrates computation, visualization, and programming in an easy-to-use environment.
- MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, and interfacing with programs written in other languages,
  including C, C++, C#, Java, Fortran and Python.
- MATLAB users come from various backgrounds of engineering, science, and economics.

https://matlab.mathworks.com/

- First of all, download and install MATLAB on your laptop:
  - https://it.unipi.it/risorse/software/matlab-mathworks/
  - https://pages.di.unipi.it/fpoloni/dida/matlab.html

Getting started with matlab



Editor
(where scripts and functions can be created and edited)

Command window
(where commands can be directly executed)

Workspace
(displays all defined variables)

See demo code (MATLAB live script):
- demo_matlab_intro.mlx

https://github.com/retico/cmepda_medphys

Intro to Matlab for Python users

https://blogs.mathworks.com/student-lounge/2021/02/19/introduction-to-matlab-for-python-users/

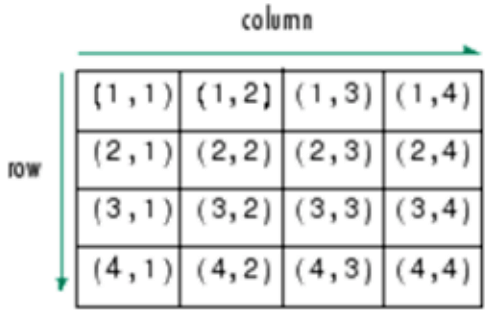
Python Syntax	MATLAB Syntax	Purpose	MATLAB Example
#	%	Comment	% hello
N/A	;	Suppress output	x(1,1);
\	...	Continue to next line	x = 1+ … 2;
os	!	Operating system command	!echo hi
+ - * /	+ - * /	Mathematical operators	x = 1+2
**	^	Exponent	x = 1^2
* / **	.* ./ .^	Element-wise operators	x = [1 2].*[3 ; 4]
not, and, or	~ & |	NOT, AND, OR logical operators	if x<2 & x>2
del	clear	Clear variable from memory	clear x y
clear	clc	Clear command window	clc

The basic data structure in MATLAB is a matrix, while Python treats everything as a general object.

https://www.mathworks.com/content/dam/mathworks/fact-sheet/matlab-for-python-users-cheat-sheet.pdf

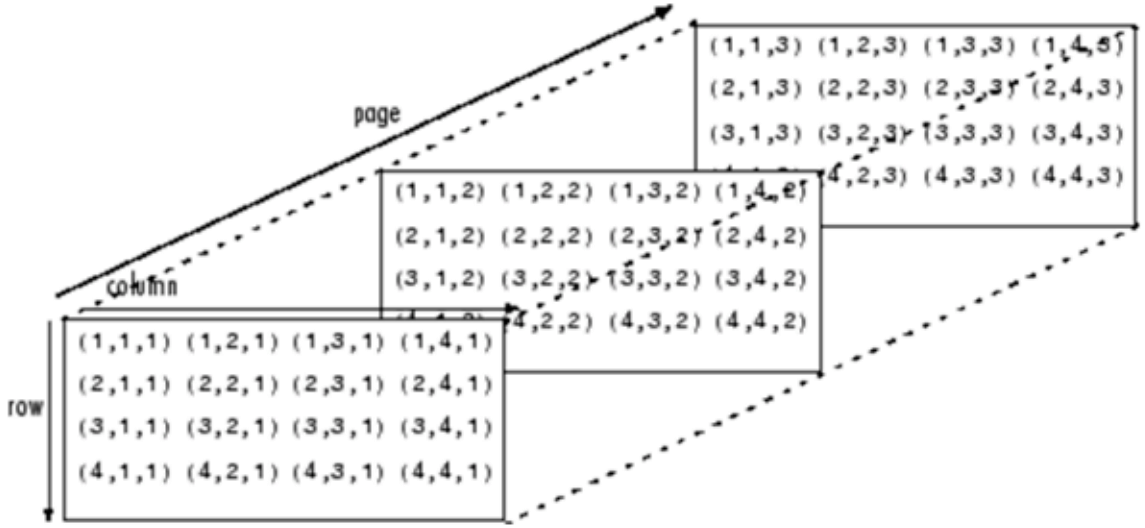
Multidimensional Matlab arrays

A multidimensional array in MATLAB® is an array with more than two dimensions. In a matrix, the two dimensions are represented by rows and columns.



In MATLAB, the basic type, is a multidimensional array. Array assignments in MATLAB are stored as 2D arrays of double precision floating point numbers, unless you specify the number of dimensions and type. Operations on the 2D instances of these arrays are modelled on matrix operations in linear algebra.

Each element is defined by two subscripts, the row index and the column index. Multidimensional arrays are an extension of 2-D matrices and use additional subscripts for indexing. A 3-D array, for example, uses three subscripts. The first two are just like a matrix but the third dimension represents *pages* or *sheets* of elements.



Data type correspondence between Python and Matlab

Python	MATLAB
float	double, single
complex	complex single, complex double
int	(u)int8, (u)int16, (u)int32, (u)int64
float(nan)	NaN
float(inf)	inf
str	str, char
bool	logical
dict	struct
list, tuple	cell
pandas.dataframe	table

Matlab m-files and user-defined functions

- Matlab programs can be written with any ASCII text editor, using the *.m file extension.
- M-files can be executed from the matlab command line.
- Try it: save the following lines of code in the plot_sin.m file and execute it

```
% plot_sin is a script that plots the sinus function
x = linspace(0,10*pi,200);
y = sin(x);
plot(y)
```

```
>> plot_sin                % it works both with and without specifying the .m extension
```

- User defined functions work just like commands in Matlab:

```
function [avg, st_dev]= show_stats(x)
% show_stats(x) returns the average (avg) and standard deviation (st_dev) of a vector (x)
% [avg, st_dev] = show_stats(x)
            n = length(x);
            avg = sum(x) / n;
            st_dev = sqrt(sum((x - avg ).^2)/(n-1));
end
```

- Try:

```
>> help show_stats
>> v1 = randn(100,1); [avg_v1,sd_v1] = show_stats(v1)
```

Create help text by inserting comments at the beginning of your program.
For a function, position the help text immediately below the function definition line.

The first help text line, often called the **H1 line**, typically includes the program name and a brief description. The *Current Folder browser* and the *help* and *lookfor functions* use the H1 line to display information about the program.

Matlab & git

- The best place to share your MATLAB projects is File Exchange because of its popularity with the MATLAB user community
- GitHub is one of the most popular websites that host Git repositories
- Since R2014b File Exchange is integrated with GitHub

- In addition to what you already know about GitHub usage, you have to register your MATLAB file extensions such as .mlx, .mat, .fig, … as binary formats. You can follow the instructions to **Register Binary Files with Git:**
  - **https://it.mathworks.com/help/matlab/matlab_prog/set-up-git-source-control.html**

You have to add some lines in the .gitattributes hidden file in your git repository

The .gitattributes file is already correctly set in the https://github.com/retico/cmepda_medphys repository

Add these lines to the .gitattributes file:

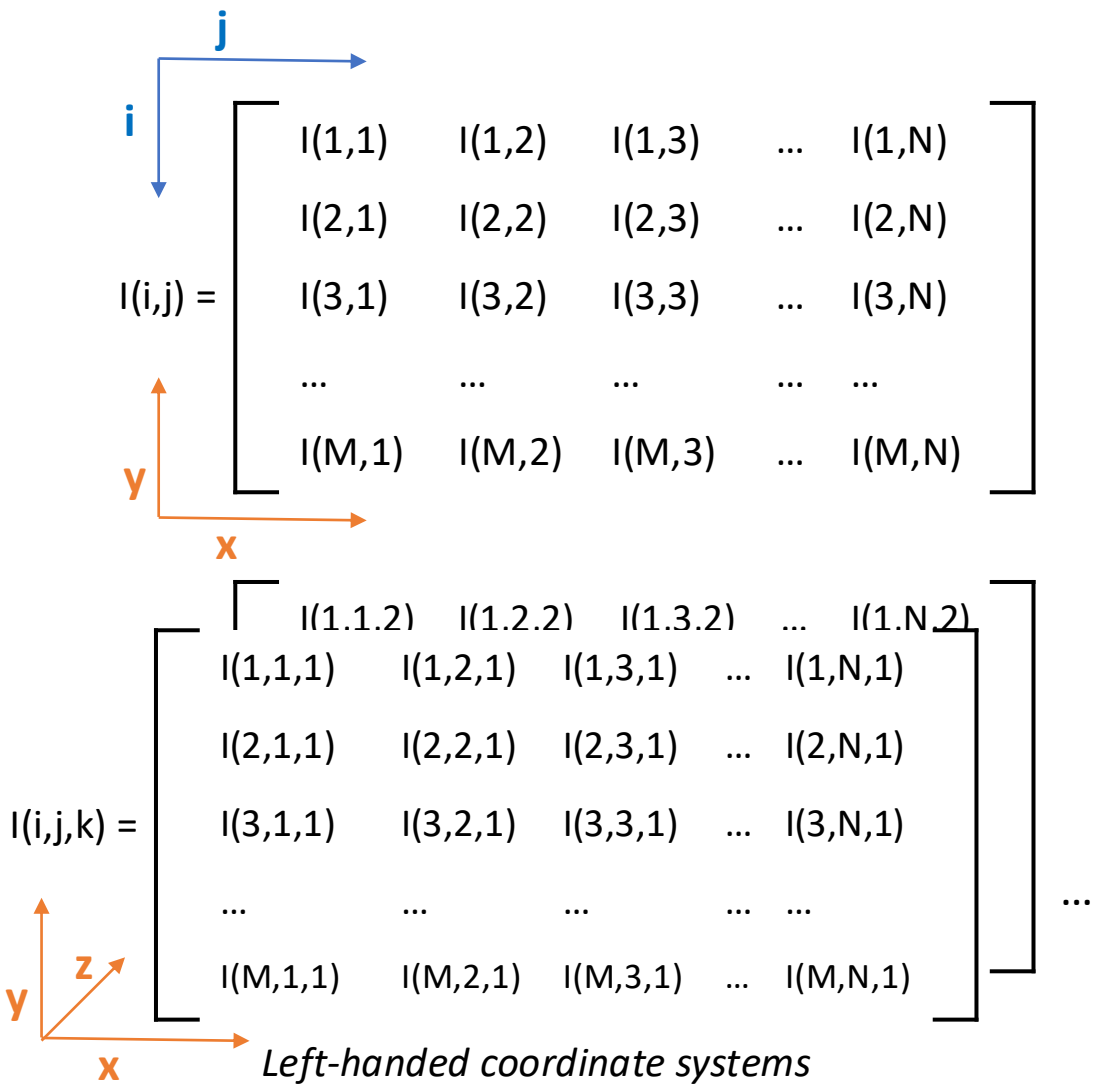
..
*.fig binary
*.mat binary
*.mlx binary
…

Basic image processing

- Basic operations with images:
  - imread(), imshow(), imwrite()
  - Histogram operations
  - Morphological Operations
  - Image filtering
  - Contrast (C), noise (N), signal-to-noise (SNR) and contrast-to-noise (CNR) ratio
  - …

- A large variety of functions for image processing are available in the MATLAB Image Processing toolbox

See demo code:
- demo_image_read_display.mlx
- demo_read_dicom.m



Left-handed coordinate systems

Hands-on materials

On GitHub https://github.com/retico/cmepda_medphys

- Read and display 3D data (DICOM dir)
  - exercise1.m
- Read and display 4D data (NIfTI file)
  - exercise2.m
- Visualization of image overlays
  - exercise3.m

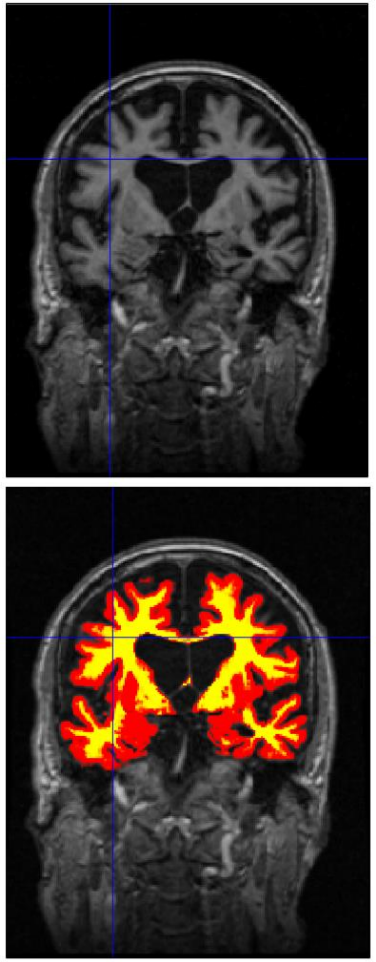


Image display and image overlay in Mango:

- Visualize and make an overlay of the segmented brain tissues on a brain MRI with data on DATASETS/IMAGES/NIfTI_Examples/Brain_segment/

Check the exercise solutions on GitHub:

exercise1_solution.m, exercise2_solution.m, exercise3_solution.m

Sample DICOM/NIfTI images

- You can download the DATASETS directory, which contains a sample of publicly available medical images, from:
  - https://pandora.infn.it/public/cmepda/DATASETS
  - and on the UniPI drive (DATASET.zip or DATASET/ folder)

- Example of open repositories of medical data
  - https://openneuro.org
  - http://fcon_1000.projects.nitrc.org
  - https://nihcc.app.box.com/v/ChestXray-NIHCC
  - https://www.cancerimagingarchive.net
  - http://www.oasis-brains.org
  - http://adni.loni.usc.edu
  - https://data.mendeley.com/datasets/ywsbh3ndr8/2
  - https://github.com/sfikas/medical-imaging-datasets

- Additional data samples (medical images/image features) for hands-on sessions will be distributed during the course

References and sources

- Books
  - Digital Image Processing for Medical Applications, Geoff Dougherty
  - Handbook of Medical Image Processing and Analysis, Isaac N. Bankman
  - Image Processing and Acquisition using Python, Ravishankar Chityala & Sridevi Pudipeddi

- Sources
  - https://it.unipi.it/risorse/software/matlab-mathworks/
  - https://pages.di.unipi.it/fpoloni/dida/matlab.html
  - https://it.mathworks.com/help/matlab/getting-started-with-matlab.html
  - https://it.mathworks.com/videos/
  - https://blogs.mathworks.com/student-lounge/2021/02/19/introduction-to-matlab-for-python-users/
  - https://mas-dse.github.io/DSE200/cheat_sheets/1_python/6_2_NumPy_for_MATLAB_users.pdf
  - https://it.mathworks.com/help/matlab/external-language-interfaces.html
  - https://it.mathworks.com/help/matlab/matlab_prog/set-up-git-source-control.html
  - https://blogs.mathworks.com/community/2014/10/20/matlab-and-git/